CS24305 — THEORY (3.0 CR) | 10-PAGE MASTER REVISION | BIT MESRA | NEP SCHEME

DCCN — 10-Page Master Quick Revision

High-Yield Formula Sheet, Protocol Matrices & Top 10 BIT Mesra PYQ Solutions

⚡ 10-Page Master Quick Revision — Data Communication & Computer Networks (CS24305)

⚡ Master Formula, Protocol & Channel Capacity Cheat Sheet

Topic / Concept	Core Mathematical Formulation / Rule	Key Exam Takeaway
Nyquist Bit Rate	$\text{BitRate}_{\max} = 2B\log_2(M)$	Applies strictly to noiseless channels with M signal levels.
Shannon Channel Capacity	$C = B\log_2(1+\text{SNR}), \quad \text{SNR} = 10^{\frac{\text{SNR}_{\text{dB}}}{10}}$	Theoretical upper bound for noisy thermal Gaussian channels.
CSMA/CD Minimum Frame	$L_{\min} = 2 \times B \times \frac{d}{v}$	Guarantees collision detection before frame transmission ends (64 bytes in Ethernet).
Stop-and-Wait Efficiency	$\eta = \frac{1}{1+2a}, \quad a = \frac{T_p}{T_t} = \frac{d/v}{L/B}$	Becomes very low over high-latency satellite channels ($a \gg 1$).
GBN vs SR Window Sizes	$\text{GBN: } W_s \leq 2^m - 1, \quad \text{SR: } W_s \leq 2^{m-1}$	Prevents sequence number window overlap on lost ACKs.
RSA Decryption Key	$(d \times e) \equiv 1 \pmod{\phi(n)}, \quad \phi(n) = (p-1)(q-1)$	Security relies on computational difficulty of factoring $n = pq$.

🔥 Top 10 High-Yield BIT Mesra Exam Questions & Solutions

Q1. A 100 km long cable has a data rate of 100 Mbps. If propagation speed is $2 \times 10^8\ \text{m/s}$, find the minimum frame size for CSMA/CD. (6 Marks)

1. $T_p = \frac{d}{v} = \frac{100\times10^3\ \text{m}}{2\times10^8\ \text{m/s}} = 5 \times 10^{-4}\ \text{s} = 0.5\ \text{ms}.$
2. $L_{\min} = 2 \times B \times T_p = 2 \times (100 \times 10^6\ \text{bps}) \times (5 \times 10^{-4}\ \text{s}) = \mathbf{100,000}\ \text{bits} = \mathbf{12,500}\ \text{bytes}.$